

CONG XU

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RESEARCH INTERESTS

Statistical machine learning: computational statistics, combination of neural networks and traditional statistical models, data analysis in complex systems.

Statistical methods addressing problems in biological and medical research: survival analysis, longitudinal data analysis, adaptive clinical trial design.

POSITIONS

Lecturer→Teaching Associate Professor Dec 2019-Present

Department of Statistics and Data Science, Southern University of Science and Technology, Shenzhen, China

Responsibilities: teaching, research, and public service

Algorithm Researcher Oct 2017-Dec 2019

Cloud and Smart Industry Group, Tencent Technology, Shenzhen, China

Responsibilities: implementation of state-of-the-art algorithms into recommendation systems; application of natural language processing and knowledge graph to projects.

Postdoctoral Researcher Oct 2015-Oct 2017

Plevritis Lab, School of Medicine, Stanford University, Stanford, California, USA

Research focus: stochastic simulation modeling to study the impact of cancer screening and treatment strategies on cancer incidence and mortality rates. **Advisor:** Sylvia Plevritis

Senior Biostatistician Jul 2014-Sep 2015

Integrated Quantitative Sciences, Novartis Pharmaceuticals, Shanghai, China

Work responsibilities: statistical analysis of randomized controlled clinical trials.

EDUCATION

2009-2014 Ph.D., Statistics

Department of Statistics, University of California, Davis

Thesis: Semiparametric Analysis of Incomplete Survival Data **Advisor:** Jane-Ling Wang

2005-2009 B.S., Statistics

Special Class for the Gifted Young (SCGY), University of Science and Technology of China

RESEARCH PUBLICATIONS

- Meng, F., Xie, J., Sun, J., **Xu, C.**, Zeng, Y., Wang, X., Jia, T., Huang, S., Deng, Y., Hu, Y. (2025) Spreading dynamics of information on online social networks. *Proceedings of the National Academy of Science* 122(4).
- Wang, J., Zhang, Y.J., **Xu, C.**, Li, J., Sun, J., Xie, J., Feng, L., Zhou, T., Hu, Y. (2024) Reconstructing the evolution history of networked complex systems. *Nature Communications* 15(1). (co-first author)
- Natarajan, P. et al. including **Xu, C.** (2024) Surgical treatment for endometrial cancer, hysterectomy performed via minimally invasive routes compared with open surgery: a systematic review and network meta-analysis. *Cancers* 16(10).
- **Xu, C.**, Wang, X., Hu, H., Qin, H., Wang, J., Shi, J., Hu, Y. (2023) A sequential re-opening of provinces for China's zero-COVID policy. *Nature Medicine* 4(5).

- Zhu, W., Jin, C., Ma, Y., **Xu, C.** (2023) Earlier recognition of scientific excellence enhances future achievements and promotes persistence. *Journal of Informetrics* **17**(2). (corresponding author)
- Delanerolle, G. et al. including **Xu, C.** (2023) The prevalence of mental ill-health in women during pregnancy and after childbirth during the Covid-19 pandemic: a systematic review and Meta-analysis. *BMC Pregnancy and Childbirth* **23**(1).
- Yu, Y., **Xu, C.**, Zhong J., Cheung, S.H. (2022) Comparison of treatments with ordinal responses in trials with sequential monitoring and response-adaptive randomization. *Statistics in Medicine* **41**: 5061-5083. (corresponding author)
- **Xu, C.**, Hadjipantelis, P.Z. and Wang, J.L. (2020) Semiparametric joint modeling of survival and longitudinal data: The R package JSM. *Journal of Statistical Software* **93**(2) doi: 10.18637/jss.v093.i02
- Caswell-Jin J.L. et al. including **Xu, C.** (2018) Change in survival in metastatic breast cancer with treatment advances: meta-analysis and systematic review. *JNCI Cancer Spectrum* **2**(4): pky062
- Munoz, D.F., **Xu, C.** and Plevritis, S.K. (2018) A molecular subtype-specific stochastic simulation model of US breast cancer incidence, survival and mortality trends from 1975 to 2010. *Medical Decision Making* **38**: 89S-98S.
- Plevritis, S.K. et al. including **Xu, C.** (2018) Association of screening and treatment with breast cancer mortality by molecular subtype in US women, 2000 to 2012. *Journal of the American Medical Association* **319**(2): 154-164.
- Lacour, J.P. et al. including **Xu, C.** (2017) Secukinumab administration by autoinjector maintains reduction of plaque psoriasis severity over 52 weeks: results of the randomized controlled JUNCTURE trial. *Journal of the European Academy of Dermatology and Venereology* **31**(5): 847-856.
- Carey, J.R., Liedo, P., **Xu, C.**, Wang, J.L., Müller, H.G., Su, Y.R. and Vaupel, J.W. (2016) Diet shapes mortality response to trauma in old tephritid fruit flies. *PLoS One* **11**(7): e0158468
- **Xu, C.**, Baines, P.D. and Wang, J.L. (2015) Improved estimation and uncertainty quantification using Monte Carlo-based optimization algorithms. *Journal of Computational and Graphical Statistics* **24**(3): 771-791.
- **Xu, C.**, Baines, P.D. and Wang, J.L. (2014) Standard error estimation in the EM algorithm when joint modeling of survival and longitudinal data. *Biostatistics* **15**(4): 731-744.

RESEARCH PROJECTS

- National Natural Science Foundation of China – Youth Project: Survival Analysis of Multiple Data Types Incorporating Deep Learning, 2022-01-01 to 2024-12-31, Principal Investigator.
- Guangdong Provincial Department of Education – Educational Science Planning Project: Quantitative Research on the Matching between Higher Education in Science and Technology and Enterprise Demand, 2021-09-01 to 2023-08-31, Principal Investigator.
- National Natural Science Foundation of China – Research Fund for International Scientists: Holistic management of complex socio-economic systems in situations of coupled epidemic instabilities, 2024-01-01 to 2026-12-31, Participant.

TEACHING RELATED

Projects

- Department of Education of Guangdong Province – Higher Education Teaching Reform Project: A New Probability and Statistics Course for Computer Science Undergraduates, 2023-12-01 to 2026-12-01, Principal Investigator.
- Department of Education of Guangdong Province – Teaching Lab Project: Probability and Statistics Teaching Lab, 2023-12-01 to 2026-12-01, Participant.

Publications

- **Xu, C.** (2025) Quantitative Research on the Alignment Between Talent Cultivation in University Statistics Programs and Industry Demands. *Education and Teaching Forum*, **22**: 169-172.
- **Xu, C., Tian, G., Li, T.** (2024) Exploration and Practice of Ideological and Political Education in Universities Based on the Concept of "Strengthening Morality and Cultivating Talents". *The Guide of Science & Education*, **36**: 86-88.
- **Li, T., Tian, G., Xu, C.** (2024) Optimization Strategies for Teaching Probability Theory and Mathematical Statistics under the Guidance of Student Needs. *The Guide of Science & Education*, **35**: 53-55.

Courses

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| ▪ Statistical Data Analysis with SAS | ▪ Probability and Statistics for Engineering |
| ▪ Survival Analysis | ▪ Fascinating Statistics |
| ▪ Probability and Statistics | ▪ Mathematical Statistics |

AWARDS

- 2024 Outstanding Teaching Award, Southern University of Science and Technology (SUSTech)
- 2024&2025 Outstanding Individual Contributor in Student Recruitment, SUSTech
- 2022 Second Prize, Young Faculty Teaching Competition, SUSTech
- 2022 Outstanding Advisor Award, China Undergraduate Mathematical Contest in Modeling
- 2022 Outstanding Organization Award, SAS China Collegiate Data Analysis Competition

SERVICES

Academic Appointments

- Council Member, Education Statistics and Management Branch, China Association for Applied Statistics
- Standing Committee Member, Health Statistics Committee, Shenzhen Health Economics Association
- Mentor, Zhiren College, SUSTech
- Director, Consultation Center, Department of Statistics and Data Science, SUSTech
- Member, Undergraduate Teaching Committee, Department of Statistics and Data Science, SUSTech

Software Development

JSM (<https://cran.r-project.org/web/packages/JSM/index.html>), an R package for joint modeling of survival and longitudinal data.